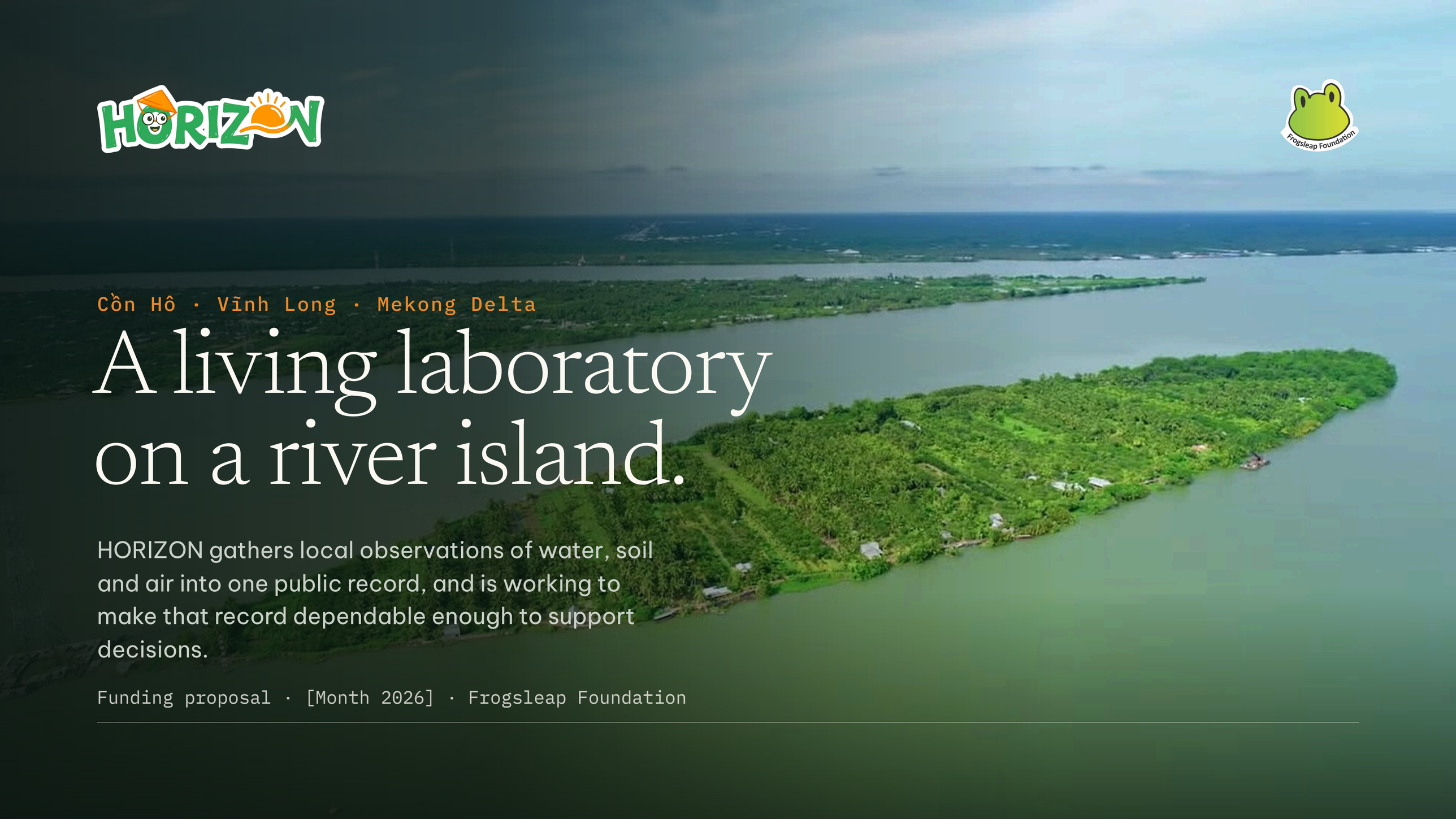


Cồn Hô · Vĩnh Long · Mekong Delta

A living laboratory on a river island.

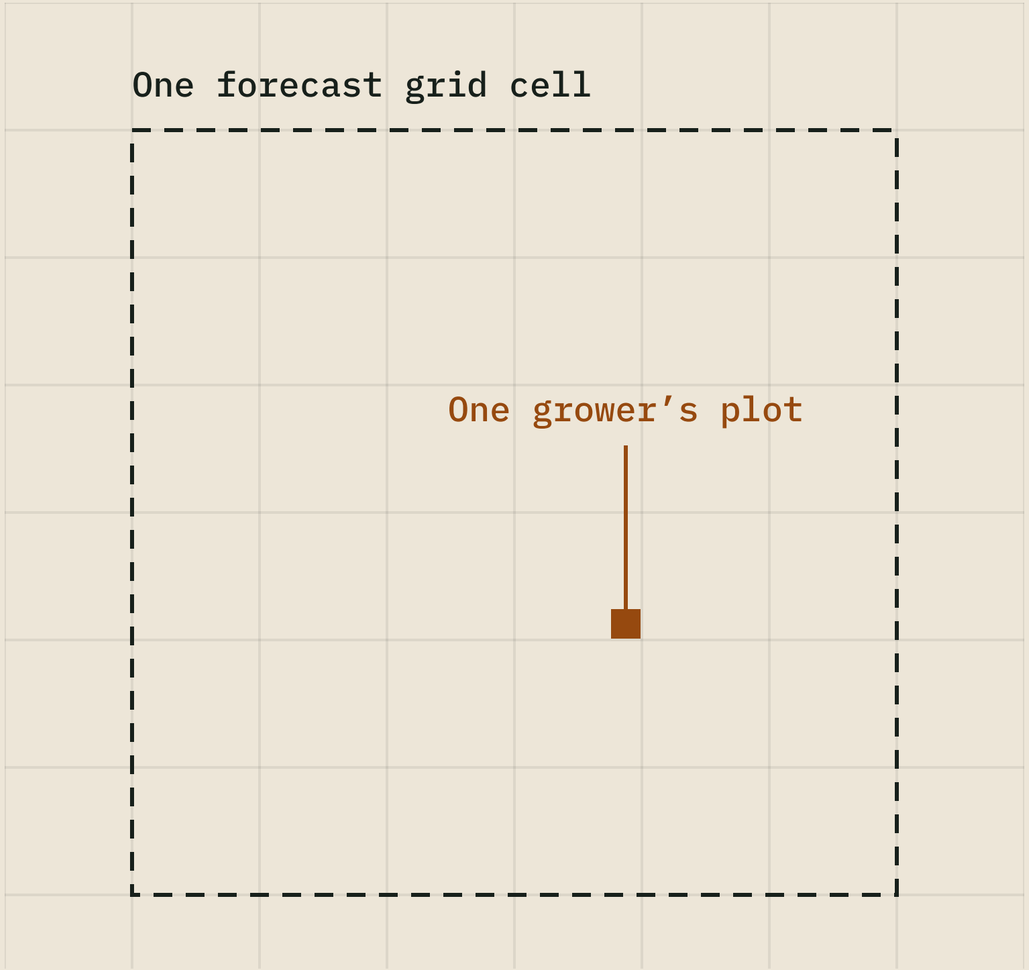
HORIZON gathers local observations of water, soil and air into one public record, and is working to make that record dependable enough to support decisions.

Funding proposal · [Month 2026] · Frogsleap Foundation

Forecasts describe a region.

Growers decide for one plot.

Whether to take river water in, irrigate or wait depends on conditions at one intake and one root zone, measured close to where the decision is made.



Schematic, not to scale



Why Cồn Hô.

A narrow island, about a kilometre long, with river water on every side. Orchards, gardens and homes sit a short walk from the bank.

Water	The river is the island's water supply. In the Delta's dry season it can turn brackish.
Soil	Moisture and salt in the root zone change with irrigation, rain and the river.
People	A small community of growers and visitors: close enough to see the sensors and use them.

Entrance to Cồn Hô · photo supplied by the project

The road to the pilot is a river.



5 Sep 2026 · river landing · the team loading pilot equipment · photo: HORIZON team

Built and checked on the island.

The station electronics were designed by the team. On site, the team assembled, checked and troubleshot them before installation.

What the team did

Designed the station boards
Wrote the firmware
Built the public website
Installed the stations in person



5 Sep 2026 · Cồn Hô · assembling station enclosures · photo: HORIZON team



Two sensing stations. One gateway.

● STATION_01

Water station

Water level, conductivity, temperature and salinity.

◐ STATION_02

Soil station

Soil moisture, EC, temperature and pH; the air above it.

No readings stored yet: battery replacement pending.

● STATION_03

Gateway

Measures nothing. Relays both stations to the cloud.

9 Oct 2026 · installed station [confirm which] · not proof of validated operation

From conditions to information.



Probes measure water, soil and air.

Radio to the gateway, then mobile data to the cloud.

Every value keeps its time and source. Missing stays missing.

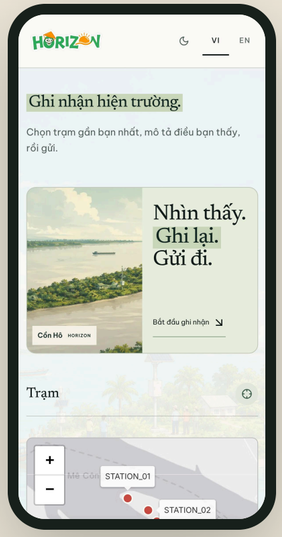
A public page shows readings and their limits.

Growers use validated information to decide.

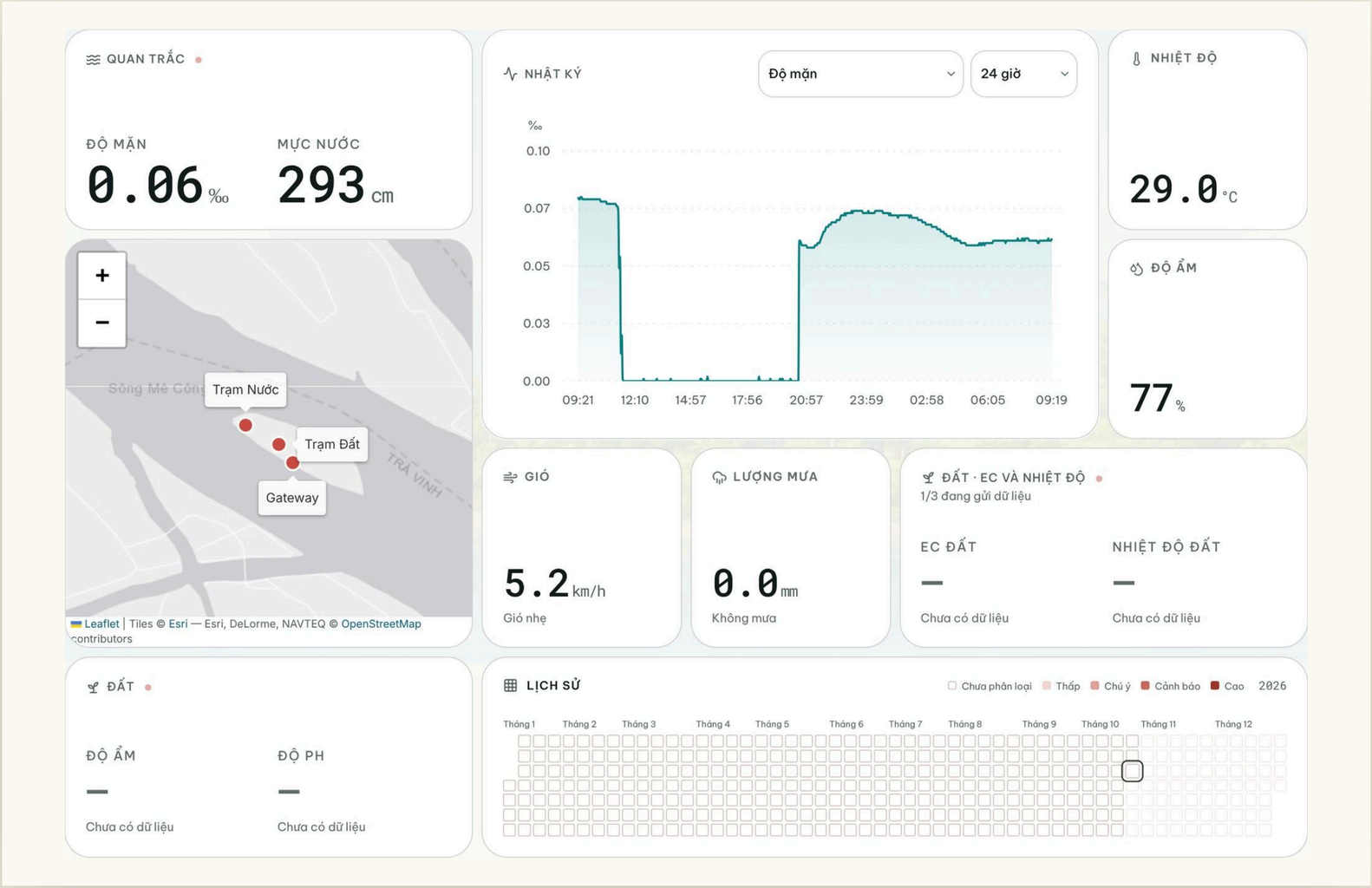
● Observed ◐ Built, not yet complete ◌ Not yet proven

One public page.

Readings, their time and their source, in Vietnamese and English. Where data is missing, the page says so.



Residents add what sensors cannot see, with a photo or a short video.



Live-site captures, 21 Oct 2026 · Vietnamese interface shown; English available

A working pilot, observed in the field.

2 + 1

sensing stations and a gateway, built by the team and taken to Cồn Hô

Field photos, 5 Sep and 9 Oct 2026

6,465

water-station readings stored in the public record since 25 Sep 2026

Counted read-only, 21 Oct 2026

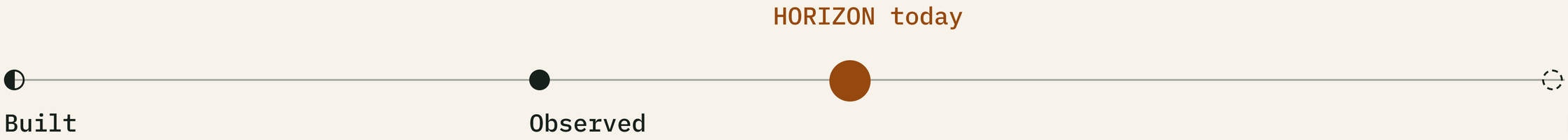
2.5 min

typical interval between new water readings

Recent rows, 21 Oct 2026

● All three observed in the field. Software test results are in the appendix.

What we have not proven yet.



Accuracy

No calibration against reference measurements yet.

Thresholds

No limit is validated for this site.

Soil record

STATION_02 has stored no readings yet.

Zero-salinity runs

Long runs of exactly 0 are under investigation.

Reliability

No long-term uptime or maintenance record.

Use

Community use and decision impact are unmeasured.

Details, causes and evidence: appendix A2 and A3

From pilot to trusted instrument.

01 **Data quality**

Explain the zero-salinity runs. Publish water conductivity and temperature. Restore soil data. [duration]

02 **Calibration**

Compare against reference measurements on site, reviewed by agronomy and hydrology advisors. [duration]

03 **Reliability**

Bench test, supervised field soak, a maintenance routine and a public uptime record. [duration]

04 **Local ownership**

Trained local caretakers, spare parts and an agreed operator for the long term. [duration]

Durations and the long-term operator: to agree with partners



What your support makes possible.

	Funding	Field partners	Science advisors
We ask for	[Amount] over [period]	Growers and cooperatives on and near Cồn Hô.	Agronomy and hydrology reviewers.
It enables	Calibration, spare parts, field visits and maintenance.	Readings tested against real irrigation decisions.	Validated methods and site thresholds.
Partners receive	Quarterly progress and an open data record.	Early access to local information, shaped by their needs.	Co-authorship of the validation work.

All amounts, periods and roles are placeholders to confirm before presenting



Help us turn readings into information people can trust.

[Name] · [email] · horizon.frogsleap.com.vn

9 Oct 2026 · Cồn Hô · photo: HORIZON team

Appendix A1

Evidence behind the main slides

Claim	Kind of evidence	Source and date
Two sensing stations and one gateway exist on Cồn Hô	Field photographs	Team photos, 5 Sep and 9 Oct 2026
6,465 water readings stored since 25 Sep 2026	Live observation	Read-only count of the public record, 21 Oct 2026
New water readings about every 2.5 minutes	Live observation	Timestamps of recent rows, 21 Oct 2026
Missing data shown as missing, never zero	Software test + live page	Unit tests; Observatory capture, 21 Oct 2026
Web app checks: 368 unit tests, browser and staging suites	Software, isolated local staging	21 Oct 2026; not proof of production behaviour
Firmware builds for both stations and the gateway	Compile-only	PlatformIO, 21 Oct 2026; nothing flashed
Dry-season salinity in the Mekong Delta	Regional context	General knowledge; not a HORIZON measurement

Appendix A2

Open findings from our own audit

Finding	What we know	Status
Zero-salinity runs (I-23)	322 of the last 1,000 water readings store salinity as exactly 0 while flagged healthy	Open: needs a read-only admin query and a probe check; no records rewritten
Water EC and temperature not public (I-18)	Stored in one column set, read from another	Fix written and rehearsed; awaiting approval to apply
Salinity estimated from EC on read failure (I-22)	Old firmware sent EC × 0.64 as a measurement	Removed at source, compiled; not yet flashed
Soil station silent	STATION_02 has no stored readings	Battery replacement pending
Device acknowledgement of remote settings	Settings are stored, not confirmed by a device	Software-tested; hardware unverified

Appendix A3

Before HORIZON is called production-ready

Gate	What remains	Roadmap step
Hardware validation	Bench test and supervised field soak, on record	Reliability
Calibration and thresholds	Reference comparison; site-validated limits	Calibration
Database changes 035 and 036	Reviewed and rehearsed; to apply with approval	Data quality
Device credentials	Rotate every production device secret	Data quality
Uploads on real storage	Confirm large video uploads on production storage	Data quality
Network trace	Record the gateway's mobile-data path once	Reliability

Each gate maps to a step on slide 11

Appendix A4

Images used in this deck

Slide	Image	Source and status
01	Aerial view of Cồn Hô	Supplied by the project
03	Entrance gate of Cồn Hô	Supplied by the project
04	Loading equipment at the river landing (IMG_1612)	HORIZON team, 5 Sep 2026; cropped, light grade
05	Assembling station enclosures (IMG_1696)	HORIZON team, 5 Sep 2026; cropped, light grade
06, 13	Installed solar-powered station (IMG_1522)	HORIZON team, 9 Oct 2026; cropped, light grade
08	Observatory and Report pages	Live-site captures, 21 Oct 2026; unedited
11	The river at dusk	Supplied by the project

No image was generated or altered in content; people pictured: consent to confirm [ ]